

MySQL Interview Questions and Answers for Fresher Data Analysts

1. What is MySQL?

MySQL is an open-source relational database management system used to store and manage data.

2. Difference between SQL and MySQL?

SQL is a query language, while MySQL is a database system that uses SQL.

3. What is a Primary Key?

A primary key uniquely identifies each row and cannot contain NULL or duplicate values.

4. What is a Foreign Key?

A foreign key is used to establish a relationship between two tables.

5. Difference between WHERE and HAVING?

WHERE filters rows, while HAVING filters grouped data.

6. What is GROUP BY?

GROUP BY is used to group similar data in a table.

7. What is JOIN?

JOIN is used to combine data from multiple tables (INNER, LEFT, RIGHT).

8. INNER JOIN vs LEFT JOIN?

INNER JOIN returns matching records, LEFT JOIN returns all from left and matching from right.

9. What is ORDER BY?

ORDER BY sorts data in ascending or descending order.

10. What is LIMIT?

LIMIT restricts the number of rows returned.

11. What is Normalization?

Normalization organizes data to reduce redundancy.

12. What is Index?

Index improves query performance by speeding up data retrieval.

13. What is a View?

A view is a virtual table based on a query.

14. What is a Subquery?

A subquery is a query inside another query.

15. How to find duplicate records?

Use GROUP BY with HAVING COUNT(*) > 1.

16. How to remove duplicates?

Use DISTINCT keyword.

17. DELETE vs TRUNCATE vs DROP?

DELETE removes rows, TRUNCATE removes all rows, DROP deletes the table.

18. How to handle NULL values?

Use IFNULL() or COALESCE().

19. Second highest salary?

Use subquery with MAX() to find second highest salary.

20. What are aggregate functions?

Functions like COUNT(), SUM(), AVG() perform calculations on data.